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SOC-15 Roll.No:- 10

Que.1Write a program to make use of basic I/O functions using different data types.

```
#include <stdio.h>

int main() {
    int a;
    float b;
    double c;
    char ch;
    char name[50];

    // Input section
    printf("Enter an integer: ");
    scanf("%d", &a);

    printf("Enter a float value: ");
    scanf("%f", &b);

    printf("Enter a double value: ");
    scanf("%lf", &c);

    printf("Enter a single character: ");
    scanf(" %c", &ch);  // Space before %c to skip leftover
newline/whitespace

    printf("Enter a string (name): ");
    scanf("%s", name);  // No & needed since name is an array (decays
to pointer)

    // Output section
    printf("\n--- Output ---\n");
    printf("Integer entered   : %d\n", a);
    printf("Float entered     : %.2f\n", b);
    printf("Double entered    : %.2lf\n", c);
    printf("Character entered : %c\n", ch);
    printf("String entered    : %s\n", name);

    return 0;
}
```

OutPut

```
Enter an integer: 10
Enter a float value: 5.5
Enter a double value: 12.345
Enter a single character: A
Enter a string (name): Darshan

Integer entered    : 10
Float entered      : 5.50
Double entered     : 12.35
Character entered  : A
String entered     : Darshan
```

Que.2:-Write a program to calculate the area of a circle.

```c
#include <stdio.h>

#define PI 3.14159

int main() {
    float radius, area;

    printf("Enter the radius of the circle: ");
    scanf("%f", &radius);

    area = PI * radius * radius;

    printf("Area of the circle = %.2f\n", area);

    return 0;
}
```

OutPut

```
Enter the radius of the circle: 5
Area of the circle = 78.54
```

Que.5:-Write a C program to demonstrate the use of single-line and multi-line comments.

```
#include <stdio.h>

int main() {
    // This is a single-line comment
    // It explains that the program prints a welcome message

    /*
       This is a multi-line comment.
       It can span across several lines.
       Here, we declare a variable and print its value.
    */

    int num = 10;  // Single-line comment: declaring and
initializing a variable

    printf("The value of num is: %d\n", num);

    /*
       End of program.
       Comments do not affect program execution;
       they are purely for documentation purposes.
    */

    return 0;
}
```

OutPut

```
The value of num is: 10
```